

NEO: Navigating Entropy in Optimized Closed-Box Video Adversarial Attacks

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Abstract—Deep learning-based video recognition systems are vulnerable to adversarial attacks, where imperceptible perturbations can induce erroneous predictions. While recent closed-box video attack methods improve efficiency via sparse querying strategies, they still suffer from high query costs and limited success rates under strict perturbation budgets. In this paper, we propose NEO, an information-entropy-guided closed-box video adversarial attack that improves both efficiency and effectiveness. We model query-based closed-box attacks as an open-box approximation problem and analyze the information required for accurate estimation. Our analysis reveals that queries near decision boundaries are more informative and that maximizing mutual information per query is critical for efficient attacks. Guided by these insights, NEO incorporates diffusion-based perturbation initialization to rapidly approach decision boundaries and a mutual-information-driven optimization strategy to enhance estimation accuracy. Extensive experiments on four benchmark datasets demonstrate that NEO achieves significantly higher attack efficiency and stealthiness than existing methods.

Index Terms—Video recognition, adversarial example, closed-box attack, deep neural network.

I. INTRODUCTION

LEARNING-BASED computer vision applications are widely used across various domains, including several security-critical areas such as intelligent assisted driving [1], [2] and biometric identification [3], [4]. However, studies have shown that deep neural networks are susceptible to adversarial attacks [5], [6], making security vulnerabilities in deep learning models a persistent concern.

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Adversarial attacks manipulate model predictions by adding imperceptible perturbations to input data [7], which can be executed during inference and require minimal attacker capabilities. As a result, they have been extensively studied across various deep learning tasks, including image classification [8], [9], audio recognition [10], [11], and natural language processing [12], [13].

In recent years, research on adversarial attacks against video models has gained attention [14], [15], [16], [17]. Unlike image and audio data, videos have higher dimensionality and contain more information, making adversarial perturbation optimization computationally expensive. To address this, Wei et al. [14] introduced the concept of sparse attacks by perturbing only a subset of frames. Subsequent studies [17], [18], [19], [20] proposed various keyframe selection algorithms (e.g., heuristic- and reinforcement-learning-based) to improve efficiency and effectiveness. In closed-box settings, where attackers lack access to the model and cannot directly compute gradients, Jiang et al. [21] employed the Natural Evolution Strategy (NES) for gradient estimation. Later works integrated NES with video-specific characteristics, leading to optimized NES-based methods [22], [23]. By combining advanced keyframe selection and gradient estimation strategies, recent methods [20] have significantly improved the success rate of closed-box adversarial attacks.

However, existing closed-box video adversarial attacks still suffer from **low success rates and high query overhead**. Even the latest method [20] requires thousands of queries for an untargeted attack while achieving a success rate below 70% under strict perturbation constraints. Moreover, the **lack of theoretical analysis** results in unclear rationales and applicability of existing methods. For instance, as observed in Section IV, keyframes change dynamically during attacks, whereas current approaches select keyframes only once before the attack begins. These observations highlight the need for a theoretical framework to analyze existing methods and guide new optimization strategies.

To address these challenges, we propose NEO, an optimized closed-box video adversarial attack guided by information entropy. We formulate query-based closed-box attacks as an open-box approximation problem, providing a unified evaluation criterion—**how closely does a closed-box step approximate the ground-truth open-box step?** Based on this informational perspective, we analyze the closed-box attack process and derive two key insights: (1) queries near decision boundaries yield more information, and (2) maximizing mutual information per query

optimizes attack efficiency. Building on these observations, we introduce two enhancement strategies: (1) diffusion-based perturbation initialization for rapid convergence to decision boundaries and (2) mutual-information-guided optimization for improved estimation accuracy.

Our main contributions can be summarized as follows:

- 1) We propose NEO, an optimized closed-box video adversarial attack guided by information entropy. NEO establishes the relationship between keyframe selection, gradient estimation, and attack overhead, providing a unified criterion by quantifying the average information entropy obtained per query.
- 2) Based on our theoretical analysis, we introduce two key strategies: diffusion-based perturbation initialization for rapid convergence to decision boundaries and mutual-information-guided estimation to enhance entropy utilization in the closed-box attack process.
- 3) Extensive experiments on multiple datasets and video recognition models demonstrate that NEO significantly improves attack success rates while reducing query overhead.

The rest of this paper is organized as follows: Sec II summarizes the related research on video adversarial attacks and provides basic knowledge about diffusion-based image mixing. Sec III provides preliminaries and the threat model of the paper. Sec IV theoretically explores closed-box adversarial attacks from an informational perspective. Sec V presents the implementation details of NEO and Sec VI lists the evaluation results of NEO. Finally, we summarize the paper and discuss potential future work in Sec VII.

II. RELATED WORK

A. Video Adversarial Attacks

1) *Open-Box Attacks*: In the open-box setting, the attacker has full access to the target model, including its architecture, parameters, and gradients, enabling direct gradient-based optimization of adversarial perturbations. Wei et al. [14] propose the first adversarial attack approach against video recognition models. Observing that adversarial perturbations can transfer among neighboring frames, they only modify a subset of all frames, which reduces the computational complexity. Subsequently, Xu et al. [17] introduce a gradient-based keyframe selection algorithm for more effective and stealthy attacks. Some other studies explore how to enhance the transferability of video adversarial examples on different models [15], [16].

2) *Closed-Box Attacks*: In the closed-box setting, the attacker has no access to the model's internal information and can only interact with the target model through queries, making gradient estimation and query efficiency central challenges. Compared to open-box attacks, video adversarial attacks under closed-box constraints are more challenging due to limited information and high computational cost. V-BAD [21], the first closed-box video adversarial attack, initializes adversarial examples using perturbations computed on a local image classification model, and adopts Natural Evolution Strategy (NES) for closed-box gradient estimation. Follow-up studies mainly

focus on three research points: (1) effective closed-box keyframe selection, (2) patch-based attacks, and (3) optimized gradient estimation.

closed-box keyframe selection : Wei et al. propose the first sparse closed-box video adversarial attack, Heuristic Attack [18], which performs heuristic keyframe selection based on model feedback. Several later studies introduce the reinforcement-learning-based mechanism for more effective keyframe selection [19], [24], [25], [26] or Bayesian optimization [27]. Recently, Chen et al. [20] propose a plug-and-play method motivated by coresets learning, which further improves the efficiency of keyframe selection. Nevertheless, most existing sparse attacks introduce additional training or query cost, and the theoretical foundation of keyframe selection remains unexplored.

Patch-based attacks : Motivated by patch-based attacks in the image domain [28], [29], Chen et al. [30] propose BSCA Attack, which applies bullet-screen-like patches to video frames and achieves attack object by optimizing the location of patches. Subsequently, Jiang et al. [31] adopt image splicing of the original image and patches of the attack target to enhance the efficiency of patch-based attacks. However, patch-based attacks usually introduce visible modifications and are therefore less stealthy than pixel-based attacks.

Optimized gradient estimation : Several previous methods aim at optimizing the gradient estimation process by introducing different prior knowledge. For example, ME-sampler Attack [22] utilizes motion maps to guide the initialization of closed-box gradient estimation process, and GEO-Trap Attack [23] adopts geometric transformations for reducing the search space for gradient estimation.

Comparison with NEO : Different from existing closed-box video adversarial attacks that are primarily designed in an empirical manner, our method NEO provides a principled theoretical framework for understanding and improving query-based attacks. Specifically, we model query-based closed-box attacks as an approximation of open-box optimization and analyze the gradient estimation process through an information-theoretic lens. This formulation avoids the ambiguity inherent in heuristic designs and clarifies the source of estimation errors introduced by limited query feedback. From a unified perspective, we show that the accuracy of closed-box gradient estimation is governed by the mutual information between the estimated gradient and the true gradient. Guided by this insight, NEO explicitly maximizes the information obtained per query and designs an information-guided closed-box attack.

B. Diffusion Models

The core idea of diffusion models realizes data generation by gradually adding and removing noise. Ho et al. [32] first propose DDPM, which combines the diffusion process with deep neural networks to optimize the quality of generation through a weighted variational lower bound. Song et al. [33] improve the sampling speed by introducing deterministic sampling paths and implicit variational inference. Dhariwal et al. [34] demonstrate the superiority of diffusion modeling in unconditional image

synthesis by improving the model architecture and training strategy. Rombach et al. [35] relocated the diffusion process to the latent space and combined it with a self-encoder to compress the image features, which significantly reduces the computational requirements and supports high-resolution image generation.

III. BACKGROUND

A. Video Adversarial Attacks

Denote the clean video dataset as $\mathbb{X} = \{X_1, X_2, \dots, X_k\}$, where $X \in \mathbb{R}^{N \times C \times H \times W}$, and N, C, H, W represent the number of frames, color channels, frame height, and frame width, respectively. Let the corresponding label set be $\mathbb{Y} = \{y_1, y_2, \dots, y_k | y \in \mathbb{Z}^+\}$, a deep video recognition model can be represented by a parameterized mapping $\mathcal{F}_\theta : \mathbb{X} \rightarrow \mathbb{Y}$. For a clean video X which satisfies $\mathcal{F}_\theta(X) = y$, the untargeted adversarial attack aims to find a human-imperceptible perturbation δ that makes $\mathcal{F}_\theta(X + \delta) \neq y$, while the targeted attack aims to make $\mathcal{F}_\theta(X + \delta) = \hat{y}$, where $\hat{y}$ is the selected target label. Video adversarial attacks can be described as the following optimization problems:

$$\arg \max_{\delta} \tilde{\mathcal{L}}(\mathcal{F}_\theta(X + \delta), y),$$

for untargeted attacks, or

$$\arg \min_{\delta} \tilde{\mathcal{L}}(\mathcal{F}_\theta(X + \delta), \hat{y}),$$

for targeted attacks, where $\mathcal{L}$ is the recognition loss function (e.g., the Cross-Entropy loss). In open-box scenarios, attackers use gradient-based methods to solve the problems and construct adversarial examples.

B. Closed-Box Gradient Estimation

Under closed-box constraints, attackers can not obtain gradient information when performing adversarial attacks. Instead, attackers adopt different gradient estimation methods for gradient-based optimization. A widely used one is Natural Evolution Strategy (NES) [36], which can be defined as:

$$\hat{\mathbf{g}} = \frac{1}{M} \sum_{i=1}^M \mathcal{S}(\mathcal{F}_\theta, X, y, \delta_i) \cdot \delta_i,$$

where M and δ denote the population size and initialized random directions (sampled from $\mathcal{N}(0, \mathbf{1}^{N \times C \times H \times W})$), respectively, and $\mathcal{S}(\cdot)$ calculates the score of each δ , e.g., for untargeted attacks, $\mathcal{S}(\mathcal{F}_\theta, X, y, \delta_i) = \mathcal{F}_\theta^y(X) - \mathcal{F}_\theta^y(X + \delta_i)$.

After gradient estimation, attackers can perform gradient-based attacks using the Fast Gradient Sign Method (FGSM) [7] or Projected Gradient Descent (PGD) [37].

C. Diffusion-Based Image Mixing

Diffusion-based image mixing synthesizes a new image by gradually combining semantic content from multiple inputs in a shared latent space. A simple and commonly used mixing rule is linear interpolation of latent embeddings (or their CLIP

projections)

$$z_{\text{mix}} = (1 - \beta) z_0 + \beta z_t, \quad \beta \in [0, 1],$$

which approximates more sophisticated spherical interpolation (slerp) used in prior work [38]. The reverse (denoising) step is then conditioned on this mixed latent, e.g.

$$x_{t-1} = \mathcal{D}_\theta(x_t | z_{\text{mix}}),$$

so that the generated image progressively reflects the blended semantics [39], [40]. These simple formulations capture the core mechanism used for CLIP-guided mixing and motivate our diffusion-based perturbation initialization.

D. Threat Model

We consider score-based query-driven closed-box adversarial attacks against video recognition models. In general, existing query-based closed-box attacks can be categorized into probability-based, score-based, and decision-based settings, depending on the type of feedback returned by the target model. Among these, score-based access, where the model returns the predicted class along with a confidence score, is commonly adopted in practical systems and subsumes probability-based feedback as a special case. In contrast, decision-based attacks rely solely on hard labels and are more restrictive.

In this work, we focus on score-based attacks, assuming that the attacker has no access to the model architecture, parameters, or gradients, but can adaptively query the target model and observe the Top-1 predicted class and its confidence score. The attacker aims to generate imperceptible adversarial perturbations that cause misclassification under a limited query budget. We do not consider transfer-based closed-box attacks, as they rely on surrogate models rather than direct interaction with the target model and fall outside the scope of our query-based formulation.

IV. THEORETICAL ANALYSIS

Although previous studies have explored improving closed-box video adversarial attacks from different perspectives (e.g., sparse attacks [19], [20], patch-based attacks [30], [31], and optimized gradient estimation [22], [23]), the theoretical foundation remains unclear. In this section, we provide some intriguing observations (empirically and theoretically) to help better understand closed-box video attacks. **Motivating example:** Keyframes keep changing during attacks. Existing sparse closed-box attacks propose different keyframe selection strategies for finding keyframes which contribute more to model prediction [18], [19], [20], [25]. Once selected, perturbations are only added on these keyframes. However, we observe that keyframes keep changing rather than remain stable during the attack process. As depicted by Fig. 1, we monitor the attack process of CLVA Attack [20] and perform the keyframe selection after each step. As we can see, the location of keyframes keeps changing during attacks. This intriguing example motivates us to rethink the role of existing optimization methods in closed-box video adversarial attacks.

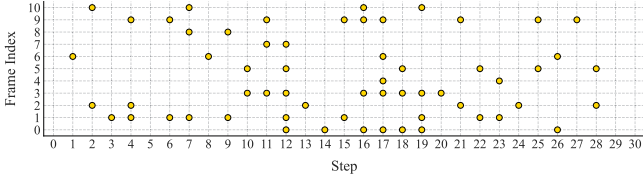


Fig. 1. A visualized example of the variance of keyframes during attacks.

To find a unifying criterion for different methods (e.g., keyframe selection, NES optimization), we consider query-based closed-box attack as an open-box approximation problem. Both settings optimize the same adversarial objective, but differ in that closed-box attacks replace the true gradient with a query-based estimator due to gradient inaccessibility. As long as the estimated gradient is aligned with the true gradient, the resulting update step approximates the corresponding open-box step. Therefore, the effectiveness of a closed-box method depends on how accurately each query captures information about the true gradient direction. Formally, this approximation can be written as

$$x_{t+1} = x_t + \alpha \hat{\nabla}_x f(x_t), \quad \hat{\nabla}_x f(x_t) \approx \nabla_x f(x_t),$$

where the approximation quality determines how close the closed-box optimization trajectory is to the open-box one. In this work, the *information about the true gradient* refers to the mutual information between the estimated gradient and the true gradient, and we later show that the magnitude of this mutual information directly determines the correctness of gradient estimation.

Based on this formulation, we can compare different methods by answering: *how similar is a closed-box step to the ground truth open-box step?* And combining the gradient estimation process of NES, this question is equivalent to: *how much information can different methods obtain about the true gradient in one query?* Guided by this informational perspective, we conduct a toy experiment on UCF-101 [41] by performing NES-based untargeted attacks on 10 randomly selected samples. The default population size is set to 40 and the step size is set to 0.01. We have the following observations additionally.

Observation 1: Queries near decision boundaries yield more information. Fig. 2(a) illustrates the variation in the average probability during the attack process, along with the change in the cosine similarity between the estimated gradient and the true gradient, compared to the initial steps of an attack, the gradient estimation precision (cosine similarity) significantly increases when the probability of the original class decreases. In other words, queries near the decision boundary can obtain more information than queries away from the decision boundary.

Therefore, the first several steps of closed-box attacks should fast approach the decision boundary. Most of existing methods utilize pixel-based image mixing to initialize attacks, so we visualized the initialization steps of different attacks in feature space. As shown in Fig. 2(b), mixing-based initialization is less effective because the shortest path in pixel space may not be the shortest path in semantic space.

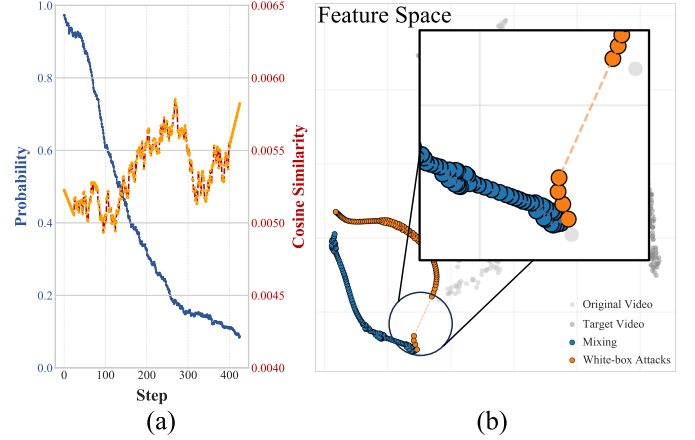


Fig. 2. (a) Relationship between Gradient Estimation Precision and Decision Boundary. (b) steps of pixel-mixing-based initialization (blue) v.s. open-box attacks (orange) in the feature space.

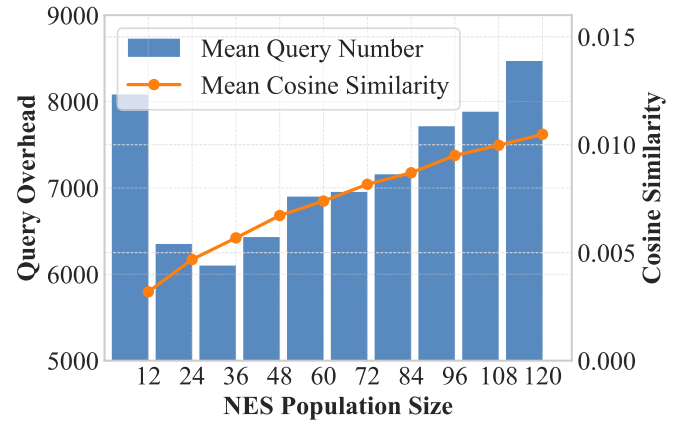


Fig. 3. Relationship between gradient estimation precision, NES population size, and query overhead.

Observation 2: Maximizing mutual information per query leads to optimal attack efficiency. We then consider the relationship between estimation precision and query overhead. As depicted by Fig. 3, with the population size of NES increases, the precision of gradient estimation also increases, which leads to fewer attack steps for success. However, the query overhead first decreases and then increases, demonstrating that the attack overhead can not be further improved by expanding the population size of NES. In other words, optimizing gradient estimation process requires improving estimation precision without increasing the query numbers.

To achieve this goal, we theoretically prove that maximizing mutual information per query leads to strictly better gradient estimation.

Theorem 1. For two closed-box gradient estimators E_1 and E_2 sharing the same evaluation model P . If E_2 achieves strictly greater mutual information with the target model than E_1 , then the estimation accuracy bounds satisfy $\mathcal{U}(E_2) > \mathcal{U}(E_1)$ and $\mathcal{L}(E_2) \geq \mathcal{L}(E_1)$, where $[\mathcal{L}(E_i), \mathcal{U}(E_i)]$ denotes the achievable accuracy interval for estimator.

Proof.

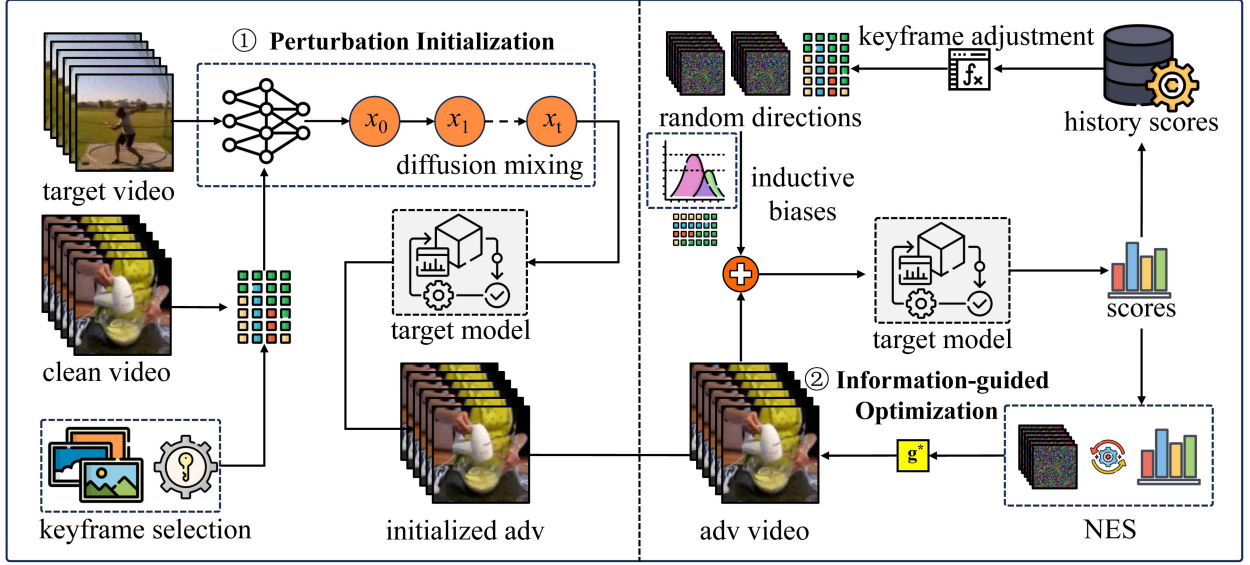


Fig. 4. A high-level overview of the workflow of NEO. For a clean video, NEO randomly selects a target video from the local dataset and performs perturbation initialization using diffusion-based image mixing. The initialized adversarial video will then be optimized using information-guided NES. During the attack process, NEO monitors the history scores, estimates the mutual information, and adaptively adjust keyframes.

Let Y be the target model's output distribution and ϵ_p the prediction error rate. By assumption, $I(Y; E_2) > I(Y; E_1)$. Through information chain rule:

$$I(Y; E_2, E_1) = I(Y; E_1) + I(Y; E_2|E_1) > I(Y; E_1).$$

For any estimator E , Fano's inequality gives:

$$H(\epsilon_p) + \epsilon_p \log(n-1) \geq H(Y|E),$$

where $H(Y|E) = H(Y) - I(Y; E)$. Let $Q(\epsilon_p) \triangleq H(\epsilon_p) + \epsilon_p \log(n-1)$, which is monotonic increasing for $\epsilon_p \in [0, 1 - \frac{1}{n}]$ when $n \geq 2$.

From Fano's inequality, we have $Q(\epsilon_p) \geq H(Y) - I(Y; E)$. Let Q^{-1} denote the inverse function. The error rate satisfies:

$$\epsilon_p \geq Q^{-1}(H(Y) - I(Y; E)).$$

Thus the accuracy upper bound becomes $\mathcal{U}(E) = 1 - Q^{-1}(H(Y) - I(Y; E))$. Since $I(Y; E_2) > I(Y; E_1)$, we immediately get:

$$\mathcal{U}(E_2) = 1 - Q^{-1}(\underbrace{H(Y) - I(Y; E_2)}_{< H(Y) - I(Y; E_1)}) > \mathcal{U}(E_1).$$

The lower bound improvement $\mathcal{L}(E_2) \geq \mathcal{L}(E_1)$ follows from the data processing inequality - additional mutual information cannot decrease the minimum achievable error. $\square$

According to the empirical and theoretical analyzes, Observation 1 and Observation 2 provide two optimization directions for closed-box video adversarial attacks: fast approach to decision boundaries and mutual information maximization.

V. METHODOLOGY

In this section, we introduce our attack framework, NEO. We first propose the overview of the workflow of NEO in Section V-A. Then we provide details of diffusion-based perturbation initialization and mutual-information-guided optimization in Section V-B and Section V-A, respectively.

A. Overview

Different from existing closed-box video adversarial attack methods, NEO sets the optimization goal of the closed-box attack process as a open-box attack and provides a unified method for evaluating the effectiveness of different methods: gradient information entropy. In other words, NEO proves that methods including keyframe selection, gradient estimation optimization, etc. are only effective when they can increase the amount of mutual information.

Motivated by the analyzes in Section IV, the distance between the current attack step and the decision boundary will significantly influence the estimation precision, while existing initialization algorithms fail to efficiently approach the boundary. Therefore, NEO introduces diffusion-based perturbation initialization to improve the effectiveness of attack initialization. Subsequently, following Theorem 1, NEO enhances mutual information per query by introducing spacial and temporal information about gradients.

Fig. 4 provides an overview of the workflow of NEO. In phase 1, for a clean video, NEO initially selects a target video from other classes (for untargeted attacks) or from the target class (for targeted attacks). Then these two video examples are fed into a diffusion model for diffusion-based semantic mixing. In perturbation initialization process, NEO performs a one-time keyframe selection, and gets an initialized adversarial example for further optimization. In phase 2, NEO optimizes the gradient

estimation process of NES by introducing spacial and temporal information in the random noises for direction searching. Specifically, NEO utilizes blurs to ensure the spacial similarity and temporal consistency of random directions. Additional, NEO stores the history attack information and adaptively adjust the keyframe following the navigating entropy to improve the attack success rate.

B. Diffusion-Based Perturbation Initialization

According to Section IV, perturbation initialization in the pixel domain is less effective because the shortest path in pixel space is probably not be the shortest path in semantic space. Therefore, a semantic-based mixing algorithm is required to fast approach to the decision boundary of the target model.

Motivated by the progress in CLIP-based image mixing using diffusion models, NEO leverages diffusion models to gradually blend the original video X with the target model X_t , driving the classifier's output toward the target class. Inspired by previous work [38], [39], NEO first encodes X and X_t into CLIP embeddings z_0 and z_t , then interpolate them in latent space:

$$z_{mix}^t = \text{slerp}(z_0, z_t, \alpha(t)), \quad \alpha(t) = \frac{t}{T}, \quad (1)$$

where $\text{slerp}(\cdot)$ ensures smooth semantic mixing. During the denoising process in Stable Diffusion, the latent variable x_t at timestep t is updated not only according to the learned diffusion model but also with an additional gradient term coming from the CLIP loss:

$$x_{t-1} = x_t - \eta(\nabla_{x_t} \log p_\theta(x_t) + \lambda \nabla_{x_t} \mathcal{L}_{\text{CLIP}}(x_t)). \quad (2)$$

In practice, NEO incrementally increases the intensity of the image mixing until the perturbation initialization is complete.

C. Mutual-Information-Guided Optimization

To enhance the average amount of information obtain per query, NEO introduces mutual-information-guided optimization. Specifically, by introducing two types of inductive bias about open-box gradients, NEO improves the estimation precision without additional query overhead.

Spatial similarity. Original NES assumes that all pixel points in a frame are independent of each other [36]. However, in practice, the covariance between pixel points shows an inverse correlation with the spatial distance in practice, which proves that the gradients of neighboring points are interrelated with each other. Thus the connection between neighboring points contains additional information about the real gradients. Therefore, we use a 2D Gaussian Filter to filter the initial random directions of NES. A 2D Gaussian Kernel can be expressed as follow:

$$G(u, v) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(u^2+v^2)/2\sigma^2}, \quad (3)$$

where u and v denote the coordinates of the points, and σ is the standard deviation. The Gaussian Filter first calculates the weights of different points in the sliding window using the Gaussian Kernel. Then the value of the center point is replaced by a weighted average of all points in the sliding window. In this

way, each point on the image establishes a connection with other points. Moreover, the strength of associations is inversely correlated with distances, which fits the natural semantic structure of video data.

Temporal consistency: Previous studies have demonstrated that adversarial perturbations can be transferred between frames [14], [18]. Meanwhile, neighboring frames are usually highly similar due to the short time interval between adjacent frames. Given that the starting point (neighboring frames) of the optimization process is similar, as well as the results (transferable perturbations), it is straightforward that the process of optimization is also similar, i.e., the gradients are transferable between neighboring frames.

Simply, we introduce this temporal consistency of gradients into the estimation process using a weighted average method. Given a n -frame initial random direction, we perform a transformation that calculates a weighted average of each neighboring frames as shown in the following equation:

$$d_{i+1} = \alpha d_{i+1} + (1 - \alpha) d_i, \quad (4)$$

where d_i and d_{i+1} denote the neighboring frames. α is the parameter to control the strength of weighted averages.

Adaptive keyframe adjustment: Instead of keeping keyframes constant, NEO adaptively adjust the keyframe following the navigating entropy. The mutual information can be calculated using the following equation:

$$I(X, Y) = \sum_{x \in X} \sum_{y \in Y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}, \quad (5)$$

where X, Y denote two random events, $p(x, y)$ denotes the joint distribution of X and Y , and $p(x), p(y)$ denote marginal distributions of X and Y , respectively. Using historical data, we empirically estimate $I(X; Y)$ by deriving joint $p^*(x, y)$ and marginal $(p^*(x), p^*(y))$ distributions from gradient samples and their evaluation scores. During closed-box attacks, if the mutual information within a time window drops below a threshold b , keyframes are re-selected to maintain attack efficacy. This adaptive process balances computational efficiency and effectiveness dynamically.

Combining diffusion-based perturbation initialization and mutual-information-guided optimization, NEO significantly improves the precision of closed-box gradient estimation without introducing additional query overhead. Algorithm 1 provides the pseudo code of NEO.

VI. EXPERIMENTAL RESULTS

In this section, we evaluate NEO on different datasets against different video classification models and compare it with previous closed-box video adversarial attacks. We first introduce the experiment setup and implementation details. Then we compare the overall performance of different attacks. Finally, we provide ablation study to further analyze NEO.

A. Experiment Setup

Datasets. We conduct experiments on three widely used action recognition datasets: UCF-101 [41], HMDB-51 [42], and

Algorithm 1. NEO Attack Framework.

Require: Clean video X , target video X_t (or target class), maximum diffusion steps T , threshold b , query limit $Q_{\max}$, and parameters η, λ, α (temporal weight), σ (Gaussian std).

Ensure: Adversarial video X_{adv} .

1: **// Phase 1: Diffusion-based Perturbation**

Initialization

2: $z_0 \leftarrow \text{Encode}(X)$ $\triangleright$ Encode clean video to CLIP embedding

3: $z_t \leftarrow \text{Encode}(X_t)$ $\triangleright$ Encode target video/class sample

4: **for** $t \leftarrow 0$ **to** T **do**

5: $\alpha(t) \leftarrow \frac{t}{T}$

6: $z_{\text{mix}}^t \leftarrow \text{slerp}(z_0, z_t, \alpha(t))$

7: Update latent variable:

$x_{t-1} \leftarrow x_t - \eta (\nabla_{x_t} \log p_\theta(x_t) + \lambda \nabla_{x_t} \mathcal{L}_{\text{CLIP}}(x_t))$

8: Select keyframes and initialize the adversarial example: $X_{\text{adv}} \leftarrow \text{Init}(x_0)$

9: **// Phase 2: Mutual-information-guided**

Optimization

10: $q \leftarrow 0$ $\triangleright$ Initialize query counter

11: **while** (attack not successful) **and** ($q < Q_{\max}$) **do**

12: **// Gradient estimation via NES**

13: Generate a set of random direction vectors $\{d_i\}$.

14: **for each** d_i **do**

15: $d_i \leftarrow \text{GaussianFilter}(d_i, \sigma)$ $\triangleright$ Enforce spatial similarity

16: **for each adjacent pair** (d_i, d_{i+1}) **do**

17: $d_{i+1} \leftarrow \alpha d_{i+1} + (1 - \alpha) d_i$ $\triangleright$ Ensure temporal consistency

18: Estimate gradient $\hat{g}$ using the modified directions.

19: Update adversarial video: $X_{\text{adv}} \leftarrow X_{\text{adv}} + \hat{g}$

20: $q \leftarrow q + 1$

21: Record gradient samples and evaluation scores.

22: Estimate mutual information $I(X; Y)$ from historical data.

23: **if** $I(X; Y) < b$ **then**

24: Adjust the keyframe in X_{adv} $\triangleright$ Adaptive keyframe adjustment

26: **return** X_{adv}

Kinetics-400 [43]. UCF-101 contains 13,320 video clips spanning 101 action categories, collected from YouTube. HMDB-51 consists of 6,766 video clips across 51 action categories, sourced from movies and online videos. Kinetics-400 is a large-scale dataset with 400 human action classes, comprising approximately 240,000 video clips.

Following previous work [20], we randomly select one video for each category from the test set of three datasets and sample them to 64 frames for classification.

Models. We evaluate our method on five deep learning models for video action recognition: LRCN [44], I3D [43], C3D [45], R3DB [46], and RNB [47]. These models have different design architectures (e.g., longterm recurrent convolutional networks,

TABLE I
CLEAN ACCURACY OF TARGET MODELS USED IN EXPERIMENTS

Models	HMDB-51	UCF-101	Kinetics-400
LRCN	32.32%	53.10%	-
C3D	79.80%	90.41%	-
I3D	81.34%	97.86%	71.80%
RNB	83.88%	97.58%	80.45%
R3DB	84.68%	98.43%	81.28%

attention mechanisms, etc.) and are widely used for evaluating previous video adversarial attacks [18], [19], [20], [25].

To ensure a fair comparison, we follow the training details in the Coreset Attack [20] to reproduce the functionality of target models. Table I lists the classification accuracies of the target models used in the experiments for clean examples, where R3DB achieves the best accuracies on three datasets. Following previous work [20], we did not train LRCN and C3D on Kinetics-400 due to limited accuracy.

For diffusion-based perturbation initialization, we adopt the pretrained Stable Diffusion V1.4 and CLIP-ViT-B32 provided by Stable Diffusion [48].

Metrics. We use 4 metrics to evaluate NEO from different dimensions (effectiveness, efficiency, and stealthiness).

- 1) Attack Success Rate (ASR): The percentage of adversarial examples that successfully mislead the target model, measuring the effectiveness of the attack.
- 2) Mean Absolute Perturbation (MAP): The average magnitude of perturbations applied to each pixel, quantifying the stealthiness of the attack. It is defined as:

$$\text{MAP} = \frac{1}{N} \sum_{i=1}^N \frac{|\delta_i|}{d},$$

where $|\delta_i|$ denotes the perturbation for the i -th adversarial example, and d is the number of pixels.

- 3) Average Queries for Success (AQS): The average number of queries required to generate a successful adversarial example, indicating the efficiency of the attack.
- 4) Average Queries for All (AQA): The average number of queries across all samples, including failed attempts, providing a broader measure of query efficiency.

Video adversarial attacks. We adopt 5 closed-box video adversarial attack methods as baseline attacks for comparison.

- 1) Heuristic Attack [18]: The first sparse closed-box video adversarial attack method which utilizes heuristic-based keyframe selection method.
- 2) RLSB [24]: Reinforcement-learning-based keyframe selection and saliency detection for critical regions.
- 3) SVA [19]: Reformulate closed-box attacks by treating the output as the environment of reinforcement learning.
- 4) AST [26]: Two-agent strategies for keyframe selection and key-region selection, respectively.
- 5) Coreset Attack [20]: Plug-and-play keyframe selection guided by K-Center-Greedy based coreset learning.

Implementation details. Our experiments are conducted on a server with two NVIDIA RTX-4090 GPUs and six Intel(R)

TABLE II
UNTARGETED ATTACK RESULTS USING FGSM+NES

Models	Attack Method	UCF-101				HMDB-51				Kinetics-400			
		ASR $\uparrow$	MAP $\downarrow$	AQS $\downarrow$	AQA $\downarrow$	ASR $\uparrow$	MAP $\downarrow$	AQS $\downarrow$	AQA $\downarrow$	ASR $\uparrow$	MAP $\downarrow$	AQS $\downarrow$	AQA $\downarrow$
LRCN	Heuristic [18] (AAAI'2020)	0.08	1.865	4153	27780	0.06	1.817	3732	28117	—	—	—	—
	RLSB [24] (IJCAI'2021)	0.15	1.871	2353	25599	0.06	1.819	4736	28126	—	—	—	—
	SVA [19] (IJCV'2022)	0.15	1.850	2030	25682	0.03	1.862	4470	28805	—	—	—	—
	AST [26] (TPAMI'2023)	0.20	1.869	4335	24752	0.24	1.821	3206	23253	—	—	—	—
	CLVA [20] (TIFS'2024)	0.21	1.834	3891	24145	0.30	1.868	3115	21032	—	—	—	—
	NEO (Ours)	0.24	1.812	2561	23241	0.35	1.814	1979	20034	—	—	—	—
C3D	Heuristic [18] (AAAI'2020)	0.23	1.816	15622	26536	0.43	1.810	9441	21042	—	—	—	—
	RLSB [24] (IJCAI'2021)	0.31	1.860	10170	23676	0.59	1.881	6062	15535	—	—	—	—
	SVA [19] (IJCV'2022)	0.32	1.823	12205	24127	0.54	1.853	7001	17218	—	—	—	—
	AST [26] (TPAMI'2023)	0.34	1.872	10871	23388	0.66	1.808	8916	15613	—	—	—	—
	CLVA [20] (TIFS'2024)	0.44	1.812	8678	20228	0.72	1.843	6457	12655	—	—	—	—
	NEO (Ours)	0.50	1.822	6941	17474	0.76	1.827	6790	11860	—	—	—	—
I3D	Heuristic [18] (AAAI'2020)	0.20	1.877	16824	27056	0.43	1.805	10988	21635	0.32	1.822	9492	23413
	RLSB [24] (IJCAI'2021)	0.37	1.861	11635	23053	0.74	1.869	6101	11879	0.44	1.847	3998	18128
	SVA [19] (IJCV'2022)	0.34	1.868	13276	24000	0.66	1.843	8940	15802	0.30	1.869	3768	21940
	AST [26] (TPAMI'2023)	0.36	1.837	12518	23411	0.69	1.852	10115	16225	0.51	1.805	8500	18596
	CLVA [20] (TIFS'2024)	0.65	1.826	10904	17460	0.81	1.813	5209	9487	0.82	1.819	3028	7442
	NEO (Ours)	0.65	1.815	7630	14811	0.85	1.815	3716	6720	0.86	1.808	3255	6061
RNB	Heuristic [18] (AAAI'2020)	0.14	1.869	11882	27284	0.22	1.834	7415	25031	0.13	1.879	13785	27398
	RLSB [24] (IJCAI'2021)	0.17	1.861	10380	26190	0.36	1.868	5644	20830	0.31	1.822	11263	24188
	SVA [19] (IJCV'2022)	0.19	1.872	8455	25638	0.23	1.852	6437	24411	0.35	1.837	10477	22909
	AST [26] (TPAMI'2023)	0.21	1.855	9316	25419	0.57	1.861	11080	19037	0.39	1.826	8838	21464
	CLVA [20] (TIFS'2024)	0.31	1.823	8544	23066	0.58	1.819	5864	15669	0.43	1.812	6224	19421
	NEO (Ours)	0.35	1.816	6525	21051	0.64	1.822	6068	14129	0.45	1.813	4352	18009
R3DB	Heuristic [18] (AAAI'2020)	0.25	1.860	13143	25723	0.22	1.861	14753	26188	0.14	1.837	6983	26338
	RLSB [24] (IJCAI'2021)	0.50	1.835	7251	18323	0.47	1.867	7089	18933	0.17	1.839	7293	26055
	SVA [19] (IJCV'2022)	0.34	1.849	10100	23023	0.30	1.825	10117	23936	0.22	1.863	4772	24446
	AST [26] (TPAMI'2023)	0.47	1.847	7318	18910	0.58	1.821	6739	16122	0.44	1.829	9865	21108
	CLVA [20] (TIFS'2024)	0.55	1.822	5643	16147	0.59	1.819	6072	15848	0.48	1.817	8894	19395
	NEO (Ours)	0.59	1.816	3587	14314	0.62	1.811	4046	12953	0.52	1.820	7059	17438

Xeon(R) Silver 4210R CPUs. The main software versions include CUDA 12.0, Python 3.9.5, PyTorch 2.0.1, etc. The values of hyperparameters used in NEO are set as follows:

- 1) Diffusion-based Initialization: We set the noise strength as 0.1, the start slerp latent style strength as 0.1, the strength step size as 0.1, and the number of diffusion steps as 50 for perturbation initialization.
- 2) Mutual-information-guided Optimization: For NES-based gradient estimation, we set the population size as 24, the learning size interval as [0.001, 0.01], the updating step size as 0.01. For untargeted attacks and targeted attacks, the maximum query numbers are set as 30,000 and 60,000, respectively. For spatial and temporal information, the kernel size and variance of the Gaussian blur are set as (3,3) and (1.5,1.5), the temporal average strength $\alpha = 0.5$.
- 3) Adaptive Keyframe Adjustment: We set the initial number of keyframes as 6, the monitoring window size as 1,000 queries, and the information boundary $b = 0.2$. When the estimated mutual information in the monitoring window is less than the boundary, NEO retains the current perturbation and re-selects 6 new keyframes.

B. Experimental Results

Overall performance comparison. Initially, we compare NEO with five baseline closed-box video adversarial attacks using FGSM and NES. Following previous work [20], we set

the length of the keyframe list as 6. In this scenario, adversarial perturbations do not accumulate iterations, thus NEO does not adaptively adjust keyframes like baseline methods. Therefore, these results show the effectiveness of different initialization and gradient estimation methods.

Table II lists the untargeted attack results using FGSM and NES. As depicted by the table, NEO achieves the highest ASR across most model-dataset combinations, demonstrating the effectiveness of diffusion-based perturbation initialization and mutual-information-guided optimization. Considering the attack efficiency, NEO significantly reduces AQS in most settings. For example, on R3DB, NEO reduces AQS by 36% and 33% compared to CLVA on UCF-101 and HMDB-51, respectively. Meanwhile, although NEO has a higher AQS than some baseline methods in several cases (e.g., on R3DB-Kinetics-400), it achieves the lowest AQA for all settings because its higher ASR. Additionally, NEO has comparable or lower MAP on different models, indicating that information-based criterion can be used to effectively evaluate and guide closed-box adversarial video attacks. Meanwhile, NEO maintains a steady improvement on state-of-the-art video classification models (e.g., RNB and R3DB) and large-scale video datasets (Kinetics-400), proving its effectiveness in cross-model transferability and real-world data scenarios.

Table III provides the targeted attack results using FGSM and NES. Compared to untargeted attacks, targeted attacks are

TABLE III
TARGETED ATTACK RESULTS USING FGSM+NES

Models	Attack Method	UCF-101				HMDB-51				Kinetics-400			
		ASR $\uparrow$	MAP $\downarrow$	AQS $\downarrow$	AQA $\downarrow$	ASR $\uparrow$	MAP $\downarrow$	AQS $\downarrow$	AQA $\downarrow$	ASR $\uparrow$	MAP $\downarrow$	AQS $\downarrow$	AQA $\downarrow$
I3D	Heuristic [18] (AAAI'2020)	0.17	2.872	59931	59086	0.36	2.717	49931	55855	0.20	2.789	40850	55398
	RLSB [24] (IJCAI'2021)	0.29	2.749	48549	55683	0.46	2.719	38549	49341	0.31	2.762	26689	49158
	SVA [19] (IJCV'2022)	0.14	2.746	56088	58717	0.43	2.762	36088	49251	0.22	2.776	25654	52269
	AST [26] (TPAMI'2023)	0.35	2.769	51647	56940	0.54	2.621	31647	44455	0.43	2.764	30282	47051
	CLVA [20] (TIFS'2024)	0.46	2.717	45272	52887	0.56	2.668	25272	39631	0.51	2.766	33901	46160
	NEO (Ours)	0.50	2.719	36614	47722	0.58	2.633	20149	36399	0.56	2.758	25824	40188
R3DB	Heuristic [18] (AAAI'2020)	0.11	2.549	56480	59339	0.13	2.512	59126	59842	0.12	2.564	29608	55560
	RLSB [24] (IJCAI'2021)	0.30	2.446	30807	50434	0.30	2.481	48907	55673	0.12	2.498	20343	54392
	SVA [19] (IJCV'2022)	0.26	2.533	45355	55760	0.24	2.555	46458	56141	0.13	2.512	27241	55293
	AST [26] (TPAMI'2023)	0.32	2.449	46160	55095	0.36	2.524	35141	50076	0.22	2.499	25189	51444
	CLVA [20] (TIFS'2024)	0.34	2.431	25102	47425	0.32	2.442	31257	50006	0.30	2.488	22912	48368
	NEO (Ours)	0.33	2.422	16880	45508	0.39	2.446	30573	47628	0.34	2.475	15411	44710

TABLE IV
UNTARGETED ATTACK RESULTS USING PGD+NES

Models	Attack Method	UCF-101				HMDB-51				Kinetics-400			
		ASR $\uparrow$	MAP $\downarrow$	AQS $\downarrow$	AQA $\downarrow$	ASR $\uparrow$	MAP $\downarrow$	AQS $\downarrow$	AQA $\downarrow$	ASR $\uparrow$	MAP $\downarrow$	AQS $\downarrow$	AQA $\downarrow$
LRCN	AST [26] (TPAMI'2023)	0.72	0.592	10503	15282	0.84	0.566	9352	11703	—	—	—	—
	CLVA [20] (TIFS'2024)	0.66	0.581	6724	13840	0.74	0.515	4504	10624	—	—	—	—
	NEO (Ours)	0.78	0.681	8323	12176	0.86	0.695	6625	9127	—	—	—	—
C3D	AST [26] (TPAMI'2023)	0.70	2.050	8436	14672	0.86	0.582	8239	10720	—	—	—	—
	CLVA [20] (TIFS'2024)	0.64	0.663	7287	15114	0.88	0.510	4410	6951	—	—	—	—
	NEO (Ours)	0.72	0.734	7069	13075	0.92	0.501	3967	5428	—	—	—	—
I3D	AST [26] (TPAMI'2023)	0.74	0.622	8359	13423	0.79	0.570	5216	10212	0.74	0.574	6459	11896
	CLVA [20] (TIFS'2024)	0.85	0.650	6726	10042	0.84	0.546	2596	6582	0.82	0.531	2218	6855
	NEO (Ours)	0.90	0.746	5749	7347	0.87	0.673	2019	4726	0.87	0.632	1908	5111
RNB	AST [26] (TPAMI'2023)	0.44	2.332	11471	21028	0.82	1.931	8118	11202	0.73	2.196	7331	13451
	CLVA [20] (TIFS'2024)	0.42	0.653	10360	21122	0.76	0.510	5325	10397	0.76	0.671	7166	11760
	NEO (Ours)	0.50	0.744	8042	18146	0.80	0.745	4305	8505	0.83	0.822	6922	9905
R3DB	AST [26] (TPAMI'2023)	0.88	1.643	7321	9319	0.90	0.557	4683	6688	0.81	1.640	5639	10002
	CLVA [20] (TIFS'2024)	0.86	0.611	5474	8201	0.89	0.518	4059	6790	0.78	0.649	3908	9407
	NEO (Ours)	0.92	0.645	4232	5966	0.93	0.560	3408	5001	0.86	0.675	3398	6159

TABLE V
TARGETED ATTACK RESULTS USING PGD+NES

Models	Attack Method	UCF-101				HMDB-51				Kinetics-400			
		ASR $\uparrow$	MAP $\downarrow$	AQS $\downarrow$	AQA $\downarrow$	ASR $\uparrow$	MAP $\downarrow$	AQS $\downarrow$	AQA $\downarrow$	ASR $\uparrow$	MAP $\downarrow$	AQS $\downarrow$	AQA $\downarrow$
I3D	AST [26] (TPAMI'2023)	0.72	2.661	25313	34495	0.75	2.776	20642	30089	0.69	2.614	20881	32700
	CLVA [20] (TIFS'2024)	0.78	2.432	21269	22747	0.82	1.940	15656	17273	0.80	2.443	22535	23835
	NEO (Ours)	0.83	2.850	16684	18254	0.84	2.297	11420	13494	0.88	2.879	14372	15338
R3DB	AST [26] (TPAMI'2023)	0.80	2.573	20901	28610	0.86	2.634	27034	30807	0.75	2.666	18448	28612
	CLVA [20] (TIFS'2024)	0.82	2.428	18657	20090	0.82	2.085	19168	20136	0.73	2.649	14859	18550
	NEO (Ours)	0.88	2.815	14905	16071	0.87	2.356	14517	16349	0.80	2.764	14690	16971

more challenging because the perturbation sizes are large and are difficult to optimize. According to Table III, similar trends can be observed as in untargeted attacks, NEO significantly improves the attack efficiency while maintaining better or comparable success rates and stealthiness. However, the magnitude of improvement is smaller compared to that in untargeted attacks, for example, NEO only reduces AQA by 4%, 5% and 8% on the R3DB models trained on three different datasets. This

is because the attack success rates of FGSM are suboptimal when performing targeted attacks and can not fully utilize the capabilities of NEO. Therefore, we further compare NEO with AST and CLVA using PGD.

Table IV and Table V provide the untargeted attack results and targeted attack results using PGD, respectively. By introducing adaptive keyframe adjustment strategy, NEO improves the ASR by 2%~10% in 12 out of 13 dataset-model combinations for

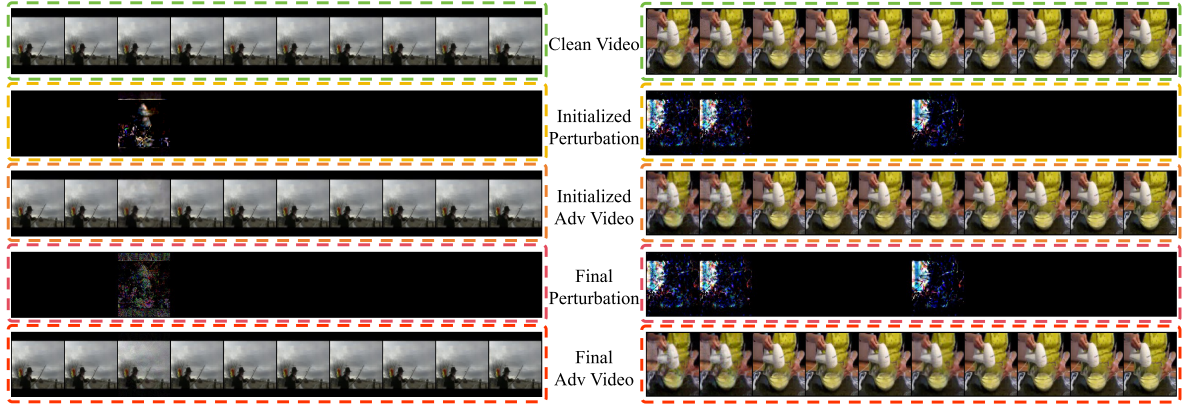


Fig. 5. Visualized examples of the untargeted attack results and the targeted attack results of NEO on the R3DB trained on UCF-101. These two videos are randomly selected from UCF-101. The original labels of the left and the right videos are 2 and 58, respectively. After attacks, the adversarial labels are 96 and 38, the MAPs are 0.635 and 2.841, respectively.

untargeted attacks. It also has 2%~8% higher ASRs for targeted attacks. It also substantially improves attack efficiency by reducing AQS and AQA. Compared to CLVA, NEO has slightly higher MAPs when utilizing PGD as optimization method, this is because for some of the examples where the CLVA attack is difficult to succeed, NEO improves the success rate of the attack by adaptively adjusting keyframes, which will correspondingly increase the MAPs. However, due to the constant perturbation size boundaries on an individual frame, NEO still maintains a high level of stealthiness.

Fig. 5 provides two visualized examples for PGD-based untargeted attacks and targeted attacks, respectively. As depicted by the figure, although NEO adds perturbations on more frames in the target attack, it is still smaller than the perturbation boundary of a single frame, making it imperceptible for manual inspection.

C. Ablation Study

To further analyze the effectiveness of the NEO components, we conducted three ablation experiments for diffusion-based perturbation initialization, inductive biases enhanced gradient estimation, and adaptive keyframe adjustment.

Initially, we evaluate how different initialization method influence the attack efficiency. As depicted by Fig. 6, we compare the gradient estimation accuracy of the first several attack steps of random initialization, pixel-mixing initialization, and diffusion-based initialization. We can observe that compared to the other two initialization methods, diffusion-based initialization effectively improves the cosine similarity between the estimated gradients and the ground truth gradients, which also proves that semantic-based mixing can help fast approach to the decision boundary.

We then explore how spatial and temporal information introduced during gradient estimation influence the estimation accuracy. Table VI lists the untargeted attack results of different combinations of NEO using FGSM. When no additional information is introduced, NEO slightly improves the attack success rate compared to CLVA due to more effective perturbation initialization, but also has a higher AQS and limited attack

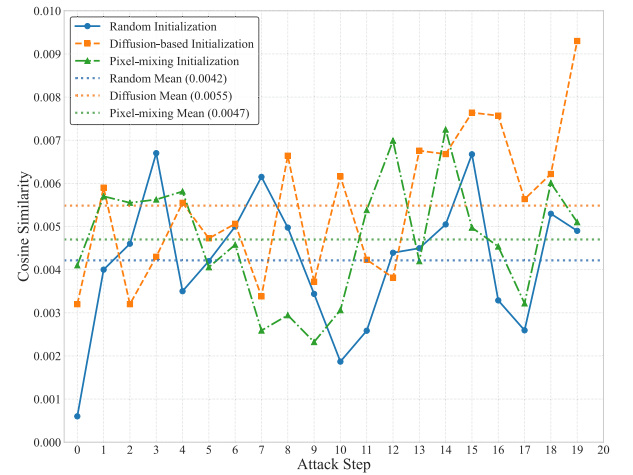


Fig. 6. Comparison of different perturbation initialization methods. Average untargeted attack results of 50 examples from UCF-101 on C3D.

TABLE VI
COMPARISON OF NEO WITH DIFFERENT INDUCTIVE BIASES

Models	Attack Method	UCF-101			
		ASR $\uparrow$	MAP $\downarrow$	AQS $\downarrow$	AQA $\downarrow$
C3D	CLVA [20]	0.44	1.812	8678	20228
	NEO w/o s&t	0.46	1.829	9772	20199
	NEO w/o s	0.46	1.824	8569	19871
	NEO w/o t	0.49	1.826	7402	18567
	NEO	0.50	1.822	6941	17474
R3DB	CLVA [20]	0.55	1.822	5643	16147
	NEO w/o s&t	0.58	1.835	6013	16022
	NEO w/o s	0.59	1.820	5578	14820
	NEO w/o t	0.59	1.819	4211	14508
	NEO	0.59	1.816	3587	14314

efficiency. By incrementally introducing temporal and spatial information, NEO further improves the attack success rate and also reduces the AQS and AQA by a large margin, showing the effectiveness of mutual-information guided gradient optimization. Furthermore, by comparing the cases where only

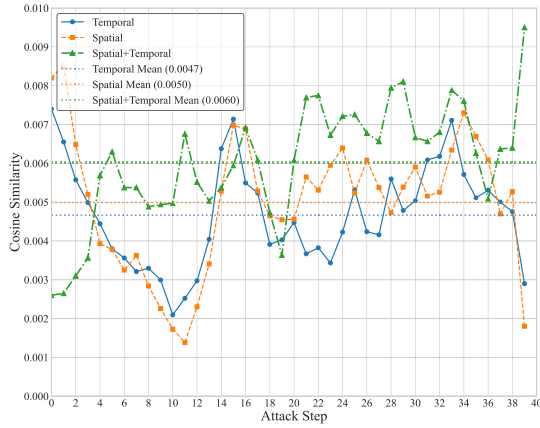


Fig. 7. Comparison of different inductive biases. Average untargeted attack results of 50 examples from UCF-101 on C3D.

TABLE VII
COMPARISON OF NEO WITH DIFFERENT KEYFRAME SELECTION STRATEGIES

Models	Keyframe Strategy	UCF-101			
		ASR $\uparrow$	MAP $\downarrow$	AQS $\downarrow$	AQA $\downarrow$
I3D	1-time	0.85	0.620	5084	8768
	each 100 steps	0.88	0.932	7072	8950
	each 500 steps	0.90	0.814	6325	7833
	adaptive	0.90	0.746	5749	7347
R3DB	1-time	0.88	0.603	5059	7882
	each 100 steps	0.92	0.951	7218	8403
	each 500 steps	0.89	0.879	4637	7093
	adaptive	0.92	0.645	4232	5966

one type of information is introduced, it can be observed that spatial information is more effective than temporal information. This is because the keyframe selection reduces the range of influence of the temporal information, while the discontinuity of perturbations also limits its effectiveness.

Fig. 7 provides the visualized results of the estimation precision with different inductive biases. As depicted by the figure, by incrementally introducing temporal and spatial information, NEO improves the precision of closed-box gradient estimation, which is consistent with the results in Table VI.

At last, we compare different keyframe selection strategies and demonstrate the effectiveness of adaptive keyframe adjustment guided by information entropy. Table VII provides the untargeted attack results of NEO using PGD and with different keyframe selection strategies. Compared to performing keyframe selection only once, re-selecting keyframes after a fixed number of iterations improves the attack success rates. However, they also introduce additional unnecessary computational overhead and more perturbations, as they update the same in scenarios where keyframes do not need to be re-selected. By monitoring the attack process and estimate the mutual information, NEO adaptively adjusts keyframes only when necessary, which maximizes attack success while ensuring attack efficiency.

D. Adversarial Defenses

Following previous work [26], we evaluate NEO against three video adversarial defense strategies, PGD-AT [37], OUD [49],

TABLE VIII
RESULTS OF NEO ATTACK AGAINST DEFENDED C3D MODEL ON HMDB-51

Metric	No Defense	PGD-AT [37]	OUD [49]	AdvIT [50]
ASR	0.76	0.53	0.62	0.60
MAP	1.827	1.855	1.834	1.826
AQS	6790	8231	7424	6258

and AdvIT [50]. These defense methods focus on robust adversarial training, model architecture modifying, and inter-frame continuity detection. Specifically, we set the optimization steps of PGD-AT to 10 and the step size to 0.1 and follow the default settings of OUD and AdvIT defenses.

As shown in Table VIII, PGD-AT achieves the strongest defense effect, leading to the largest reduction in attack success rate. However, this robustness comes at the cost of increased training overhead and potential degradation of model utility. As a query-based closed-box attack, NEO remains effective against OUD. Moreover, since NEO optimizes perturbations with smaller magnitude during the attack process, it exhibits improved robustness against AdvIT.

Potential solutions Based on experimental results, existing video adversarial defense methods are vulnerable to advanced closed-box attacks. Meanwhile, training-based robustness enhancement methods cause unacceptable degradation in normal performance and increased costs. Therefore, designing active defense methods that do not require training (like stateful adversarial defenses for image models [51]) represents a potential solution.

VII. CONCLUSION

In this work, we propose NEO, an optimized closed-box video adversarial attack navigated by information entropy. NEO provides an informational perspective for the query-based closed-box attack by considering it as an open-box attack approximation problem. From this perspective, we rethink the role of existing optimization methods in closed-box video attacks and find two observations: (1) queries near decision boundaries yield more information and (2) maximizing mutual information per query leads to optimal attack efficiency. Following these observations, we introduce: (1) diffusion-based perturbation initialization for fast approach to the decision boundary and (2) mutual-information-guided gradient estimation for high estimation precision. We also enhance the effectiveness of the keyframe selection by introducing an information-guided adaptive adjustment strategy. Comprehensive experiments on three widely used datasets and five video recognition models show that NEO effectively improves the attack success rate and attack efficiency compared with baseline attacks, while maintaining comparable attack stealthiness.

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